

Model Development Document

Complaint → Regulation Classifier (CMPL-REG-24)
First line of defense — Model Development

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LLM: nim (meta/llama-3.1-8b-instruct)
Data: CFPB Consumer Complaint Database (real, redacted narratives)
Dataset: 4000 rows · regulatory rate 0.875
Repo: reg-agents · model id: CMPL-REG-24

1 · Executive summary & methodology

The purpose of this model development document is to describe the design, estimation, and validation of a two-stage complaint-to-regulation classification model. This model is intended to classify consumer complaints into one of several regulatory categories, with the ultimate goal of informing regulatory actions.

The materiality of this model is significant, as it will be used to inform regulatory decisions that have a direct impact on consumers and financial institutions. The model's accuracy and reliability are therefore critical to ensuring that regulatory actions are fair, effective, and compliant with relevant laws and regulations.

The design rationale for the two-stage architecture of this model is based on an economic and statistical decision. In the first stage, a high-recall gate is used to filter out complaints that are unlikely to be related to a specific regulatory category. This gate is implemented using a TF-IDF classifier, which is a simple and computationally efficient approach that is well-suited to this task. The second stage of the model uses a more complex and computationally expensive approach, a retrieval-augmented generation pipeline (NeMo Retriever embeddings + LLM labeler served via NVIDIA NIM), to classify complaints that have passed through the first stage. This approach is more accurate, but also more computationally intensive, and is therefore only applied to complaints that are most likely to be related to a specific regulatory category.

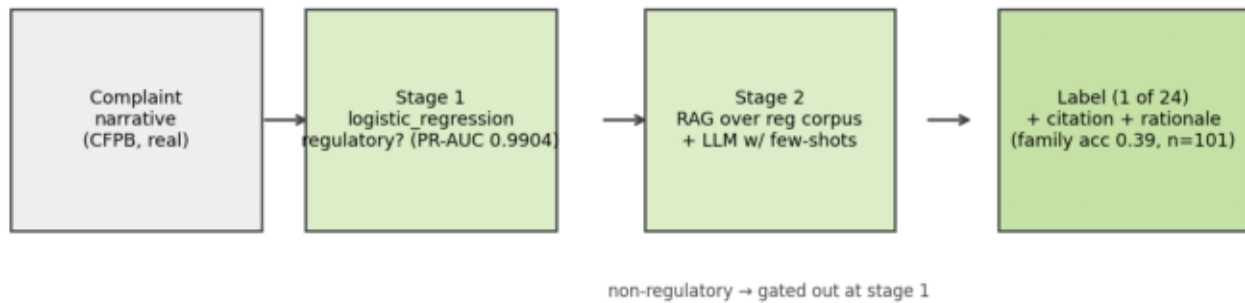
The estimation and model-selection protocol for the first stage of the model involved a bake-off between two different classifiers: logistic regression and XGBoost. The logistic regression model with L1 regularization and a regularization parameter of 4.0 was selected as the champion model, based on its high precision-recall area under the curve (PR-AUC) of 0.9936 on the validation set. The XGBoost model with 300 estimators, a maximum depth of 6, and a learning rate of 0.1 was the runner-up, with a PR-AUC of 0.988 on the validation set. The confusion matrix for the champion model is provided in the metrics section, and shows a high level of accuracy in classifying complaints into the regulatory categories.

The second stage of the model uses a retrieval-augmented generation pipeline (NeMo Retriever embeddings + LLM labeler served via NVIDIA NIM) to classify complaints that have passed through the first stage. The model was evaluated on a small set of 101 complaints, and achieved an accuracy of 0.2475, a family accuracy of 0.3861, and a macro F1 score of 0.2516. The model's performance varied across different regulatory categories, with some categories (such as BSA_AML and FCRA_ACCURACY) achieving high recall rates, while others (such as ECOA_ADVERSE_ACTION and FDCPA_THREATS) achieved low recall rates.

One known weakness of the model is the use of weak labels in the second stage. The model is trained on a small set of labeled complaints, but the labels are not always accurate or consistent. This can lead to biases in the model's performance, particularly if the weak labels are not representative of the underlying data. To mitigate this risk, the model's performance is monitored and evaluated regularly, and any biases or inconsistencies are addressed through retraining or re-labeling.

The model's performance is also monitored through a set of guardrails, which are designed to detect and prevent any biases or inconsistencies in the model's performance. These guardrails include metrics such as the PR-AUC and ROC-AUC, which are used to evaluate the model's performance on different regulatory categories. The model's performance is also monitored through a set of regular evaluations, which are designed to detect any changes or biases in the model's performance over time. If any biases or inconsistencies are detected, the model is retrained or re-labeled to address the issue.

2 · Architecture



Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

2 · Architecture (detail)

Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: `classify_complaint`, `sample_complaints`, `get_model_metrics`) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

3 · Data & curation

Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 200 reserved as a 5% scoring holdout for the batch-ingestion layer before any modeling split, then 3040 train / 380 validation / 380 test; regulatory rate 0.875).

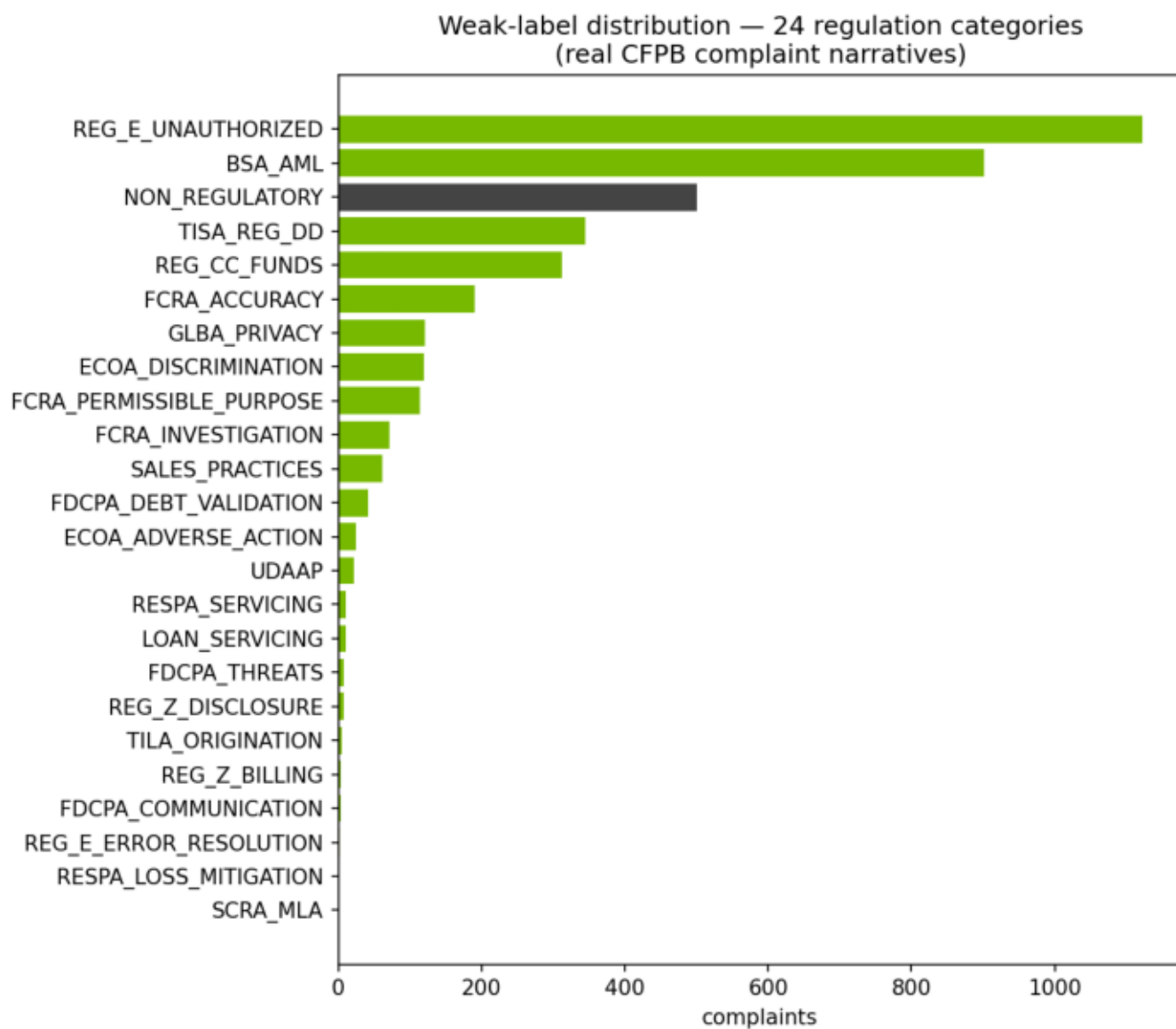
Curation (scripts/fetch_cfpb_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

3 · Dataset summary

property	value
source	CFPB Consumer Complaint Database (public, PII-redacted)
curated rows	4,000
scoring holdout (reserved first)	200 (5%, stratified — fed only through the ingestion layer)
train / val / test split	3,040 / 380 / 380 (stratified 80/10/10 of the remaining 95%)
regulatory rate	87.5%
taxonomy coverage	24 of 24 categories
curation stages	length filter · exact dedup · near dedup · PII check · balanced sampling
label provenance	weak supervision (CFPB issue taxonomy + keyword rules)

3 • Label distribution



Weak-label distribution over the 24 regulation categories.

3.1 · Regulation taxonomy (with dataset support)

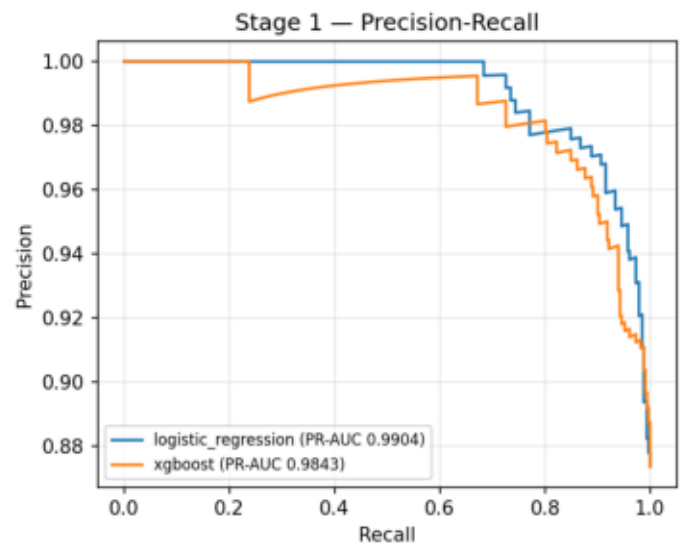
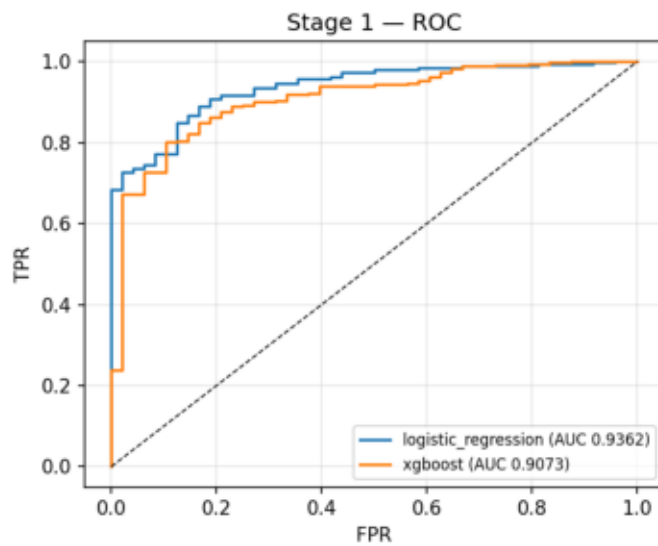
code	regulation / category	n in dataset
FCRA_ACCURACY	FCRA — Accuracy of Reported Information	191
FCRA_INVESTIGATION	FCRA — Reinvestigation of Disputes	72
FCRA_PERMISSIBLE_PURPOSE	FCRA — Permissible Purpose / Improper Use	114
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation / Not Owed	42
FDCPA_COMMUNICATION	FDCPA — Communication Tactics	3
FDCPA_THREATS	FDCPA — False Statements or Threats	8
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthorized Transfers	1122
REG_E_ERROR_RESOLUTION	Reg E / EFTA — Error Resolution	2
REG_Z_BILLING	Reg Z / FCBA — Billing Error Disputes	4
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Interest & Disclosures	8
TILA_ORIGINATION	TILA — Credit Origination & Underwriting Disclosures	5
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Discrimination	119
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse Action Notices	25
RESPA_SERVICING	RESPA — Mortgage Servicing & Escrow	11
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation & Foreclosure	1
TISA_REG_DD	TISA / Reg DD — Deposit Account Disclosures	345
REG_CC_FUNDS	Reg CC — Funds Availability	312
BSA_AML	BSA/AML — Account Freezes & Closures	901
GLBA_PRIVACY	GLBA / Privacy — Information Sharing & Safeguards	121
UDAAP	UDAAP — Unfair, Deceptive, or Abusive Acts or Practices	22
SALES_PRACTICES	Sales Practices — Unauthorized Accounts/Products	61
SCRA_MLA	SCRA / MLA — Servicemember Protection	1
LOAN_SERVICING	Loan Servicing (Auto/Student/Personal) — UDAAP & S	10
NON_REGULATORY	Non-Regulatory — General Service	500

4 • Stage-1 bake-off leaderboard

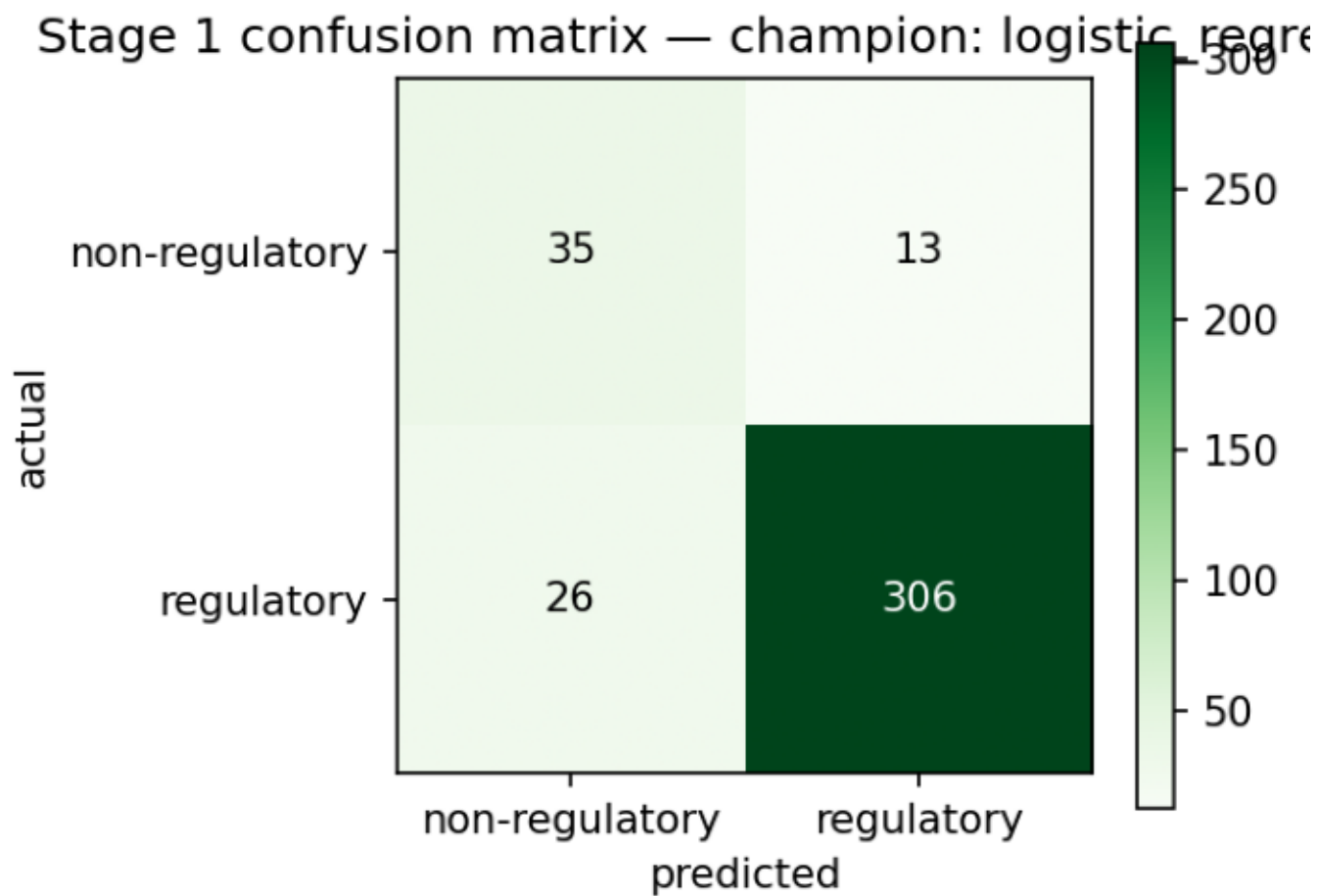
	model	params	val_pr_auc	threshold	pr_auc	train_roc_auc	roc_auc	train_test_g	f1	precision	recall	accuracy
logistic_regression		l1, C=4.0	0.9936	0.491	0.9904	0.9963	0.9362	0.0601	0.9401	0.9592	0.9217	0.8974
n_estimators=300, max_depth=3			0.988	0.545	0.9843	1.0	0.9073	0.0927	0.9412	0.9426	0.9398	0.8974

Champion: logistic_regression - selected on validation PR-AUC; other columns are one-shot test metrics.

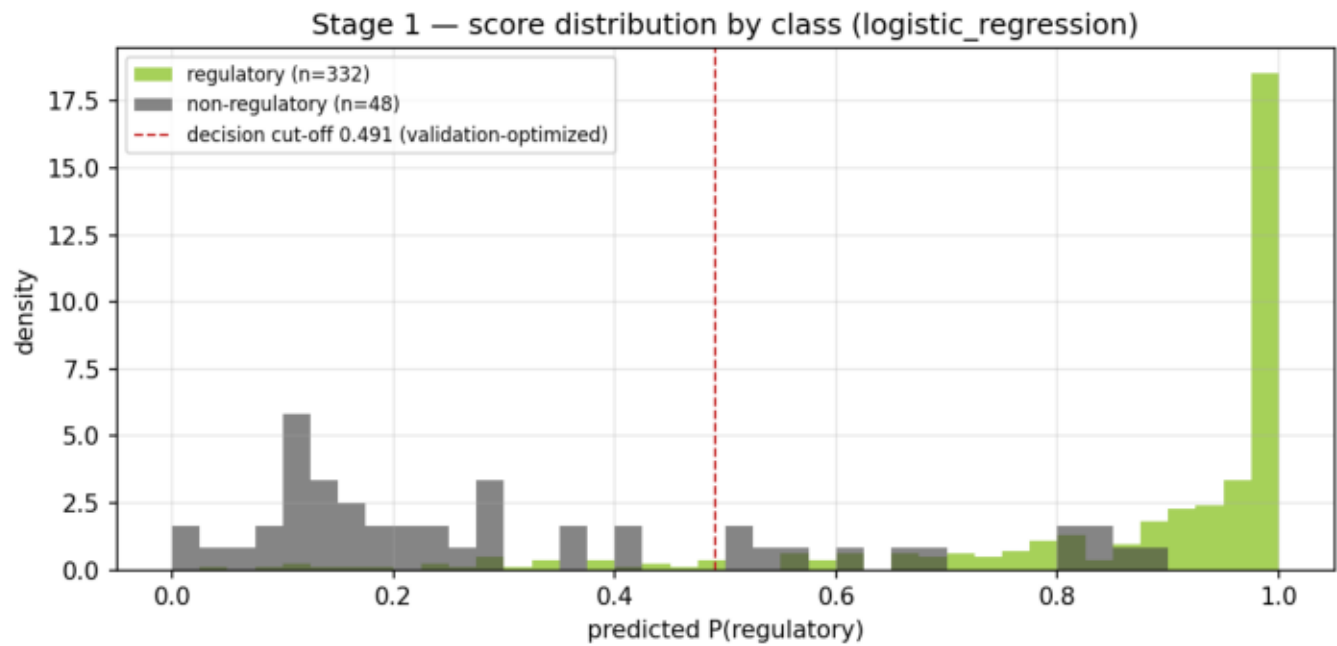
4 • Stage-1 ROC / PR curves



4 • Stage-1 confusion matrix

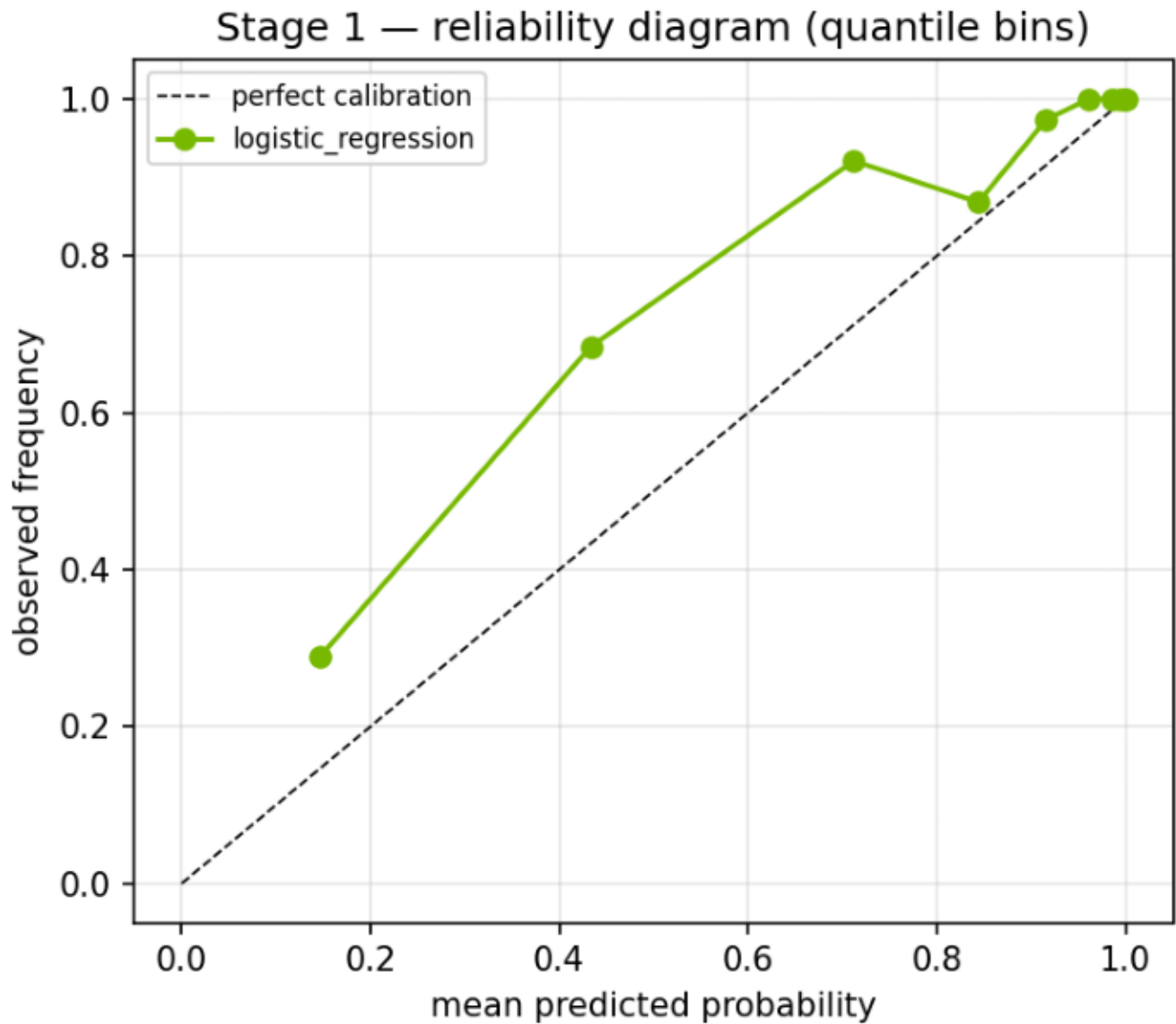


4.1 • Score separation



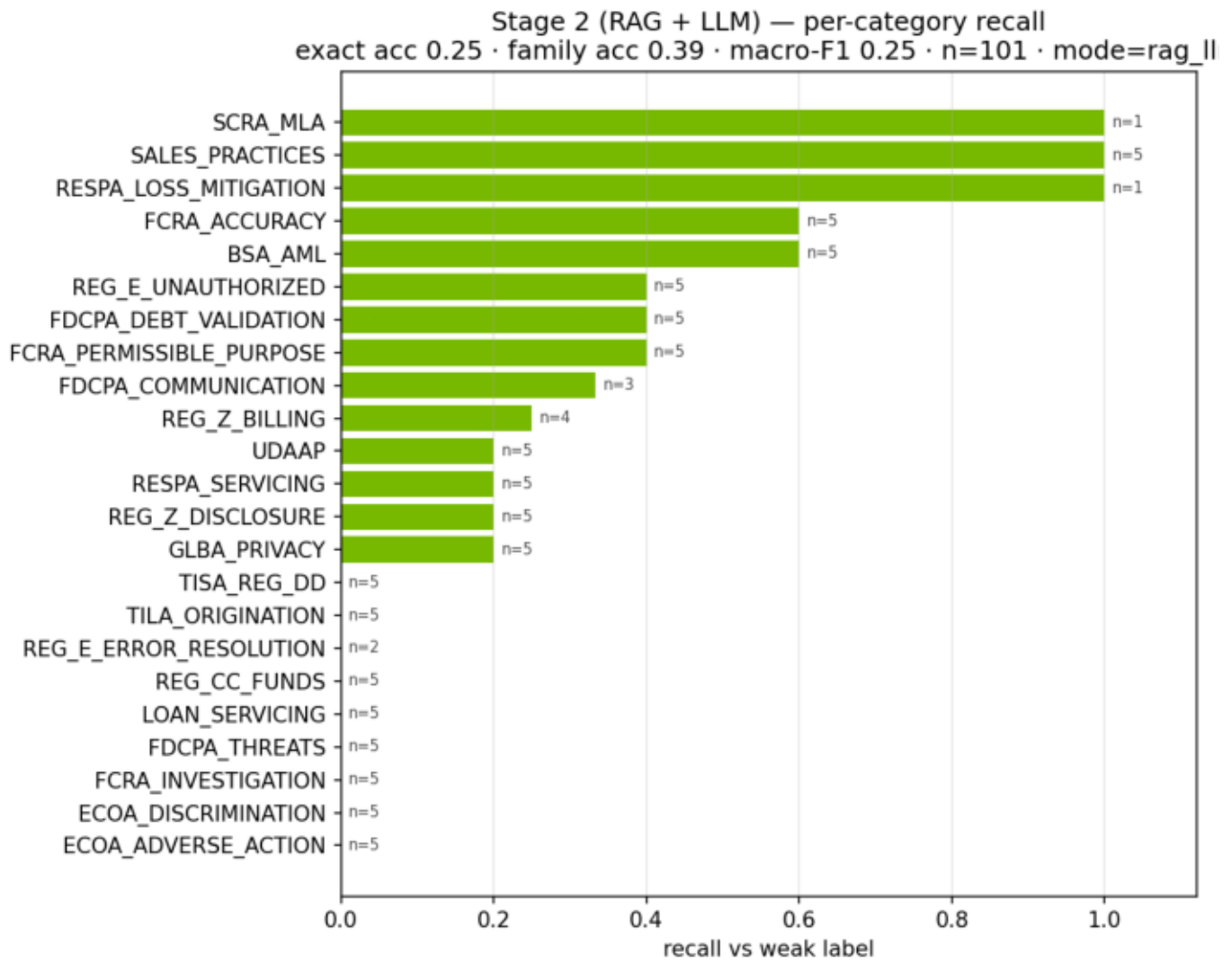
Class separation at the validation-optimized cut-off 0.491 (default 0.5 not used).

4.1 · Calibration (reliability diagram)



Whether predicted probabilities can be read as probabilities.

5 · Stage-2 per-category recall



Exact acc 0.2475 · family acc 0.3861 · macro-F1 0.2516 · n=101.

5 • Stage-2 per-category detail

category	support	recall
BSA_AML	5	0.6
ECOA_ADVERSE_ACTION	5	0.0
ECOA_DISCRIMINATION	5	0.0
FCRA_ACCURACY	5	0.6
FCRA_INVESTIGATION	5	0.0
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_DEBT_VALIDATION	5	0.4
FDCPA_THREATS	5	0.0
GLBA_PRIVACY	5	0.2
LOAN_SERVICING	5	0.0
REG_CC_FUNDS	5	0.0
REG_E_UNAUTHORIZED	5	0.4
REG_Z_DISCLOSURE	5	0.2
RESPA_SERVICING	5	0.2
SALES_PRACTICES	5	1.0
TILA_ORIGINATION	5	0.0
TISA_REG_DD	5	0.0
UDAAP	5	0.2
REG_Z_BILLING	4	0.25
FDCPA_COMMUNICATION	3	0.333
REG_E_ERROR_RESOLUTION	2	0.0
RESPA_LOSS_MITIGATION	1	1.0
SCRA_MLA	1	1.0

Recall vs weak labels on the stratified evaluation sample.

6 · Guardrail inventory

guardrail	mechanism	surfaced as
Taxonomy whitelist	LLM label must be one of the 24 codes; else re	complaint_classifications_total{label}
Strict-JSON parsing	malformed LLM output -> deterministic keyword	mode=fallback counter (alertable spike)
Stage-1 gating	non-regulatory volume never reaches the l	cost + attack-surface control
Retrieval grounding	answer must cite a retrieved corpus exce	citation attached to every prediction
Provider portability	OpenA <-> NIM via config; keyword mode if b	mode label per classification

Sign-off

Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.